Heterogeneous Effects of Minimum Wage Policies: Evidence from California Counties Using Synthetic Control Methods

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Abstract

This paper examines the heterogeneous effects of minimum wage increases across local labor markets, focusing on limited-service restaurants (NAICS 722513), full-service restaurants (NAICS 722511), and retail trade (NAICS 44-45). Using the Stacked Synthetic Control Method (SCM), we analyze California's statewide minimum wage increases implemented in 2014 Q3 and 2016 Q3, covering the period from 2009 Q1 to 2019 Q4. We find that employment and establishment growth in the limited-service restaurant sector is concentrated in large and medium counties, whereas small counties showed muted employment responses. These patterns align with monopsonistic labor market theory, highlighting differences in labor supply elasticity. Higher labor supply elasticity in large and medium counties allowed firms to expand employment and establishments after minimum wage increases, while firms in smaller counties primarily adjusted wages rather than employment. In contrast, full-service restaurants and retail trade exhibited weaker or negative responses across all county sizes, suggesting that industry-specific constraints affected adjustments. Furthermore, the delayed employment response in the fast food sector, emerging predominantly after the second wage increase in 2016, implies that firms revised expectations upon recognizing sustained policy changes. Our findings underscore the importance of labor supply elasticity and support the monopsony framework in explaining employment dynamics across heterogeneous markets.

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1 Introduction

Over the past three decades, the study of minimum wage policies has evolved through the adoption of new empirical methods and theoretical frameworks. Traditional neoclassical models predict that binding minimum wages reduce employment due to substitution and scale effects. However, seminal studies by Card (1992a), Card (1992b), and Card and Krueger (1994) challenged this view, finding little evidence of disemployment effects and, in some cases, observing positive employment impacts. These findings led to a reassessment of competitive labor market models and contributed to the emergence of the "new minimum wage literature," which applies quasi-experimental methods to assess policy impacts (Dube and Lindner, 2024).

Since then, research has increasingly emphasized imperfect competition models, particularly monopsony, to explain why minimum wage increases may not lead to job losses predicted under perfect competition. Monopsonistic labor markets are characterized by firms with wage-setting power, meaning that wages are often below the competitive equilibrium. Under these conditions, a binding minimum wage can increase both wages and employment by correcting labor market distortions. One of the most important recent empirical studies,

Wiltshire et al. (Forthcoming), provides strong evidence to support this framework, demonstrating that even substantial increases in minimum wage, including those that exceeded $15 per hour, did not result in employment declines in the limited-service restaurant industry. Instead, they found a significant increase in weekly wages, a reduction in worker separation rates, and partial price pass-through, rather than employment reductions. These findings reinforce the idea that labor market frictions and firm wage-setting power play a crucial role in shaping employment responses to wage policies.

This paper builds on Wiltshire et al. (Forthcoming) by examining a key yet often overlooked dimension: the role of the size of the local labor market in determining firm responses to minimum wage increases. While existing research has largely treated labor markets as homogeneous, we explicitly account for differences across counties by stratifying them into Large, Medium, and Small categories. This approach allows us to capture heterogeneity in firm behavior and labor market conditions, which can drive divergent employment and establishment responses.

We analyze the effects of two statewide minimum wage increases in California, using data spanning from 2009 Q1 to 2019 Q4. The first increase, implemented on July 1, 2014, raised the minimum wage from $8.00 to $9.00, while the second increase, effective January 1, 2016, further raised it to $10.00. These policy changes provide a natural setting to examine how firms in different-sized labor markets adjust over time. Using the Stacked Synthetic Control Method (SCM), we estimate the effects of minimum wage increases on employment, establishment counts, and average wages in the limited-service restaurant (NAICS 722513), full-service restaurant (NAICS 722511), and retail trade (NAICS 44-45) sectors.

This study contributes to the minimum wage literature in three key ways. First, we demonstrate that employment and establishment responses to minimum wage increases vary significantly by county size. In the limited-service restaurant sector, large and medium counties experienced positive and statistically significant employment and establishment growth, particularly after the second minimum wage increase in 2016 Q3. In contrast, small counties exhibited muted employment responses, suggesting that firms in smaller labor markets adjusted primarily through wage increases rather than hiring expansion. However, the full-service restaurant and retail trade sectors showed weaker or negative employment and estab-

lishment growth, reinforcing the idea that industry characteristics and cost structures shape firm responses to minimum wage policies.

Second, we integrate a monopsony framework to explain the observed heterogeneity in employment and establishment responses. By classifying counties according to the size of the labor market, we show that firms in smaller markets, where labor supply is more inelastic, responded differently than those in larger, more competitive labor markets. The muted employment effects in small counties are consistent with the prediction that in less elastic labor markets, firms adjust primarily through higher wages rather than increased hiring. In contrast, the expansion of employment and establishments in larger counties suggests that the minimum wage helped correct monopsony distortions, facilitating greater labor market participation and firm growth.

Third, we document a delayed response in employment and establishment growth, with significant effects emerging only after the second minimum wage increase in 2016 Q3. This suggests that firms initially viewed the first increase as a temporary policy shift but revised their expectations after the second increase, adjusting their long-term expansion strategies accordingly. To further explore the role of market structure, we introduce the percentage of chain restaurants (hereafter, chainness rate) as a proxy for market concentration within the limited-service restaurant sector. Our results show that in large counties, where chain concentration is higher, the effects of minimum wage on employment tend to be more pronounced, suggesting that firms with higher chain presence were better able to expand following wage increases.

We perform two robustness checks to verify the stability of our findings. First, we implement the Difference-in-Differences approach of Callaway and Sant'Anna (2021), which allows for heterogeneous treatment effects across units and time. Second, we modify the donor pool selection strategy by relaxing the initial stratification criteria and including a broader set of potential donor counties. Across both robustness checks, we find that the key results remain consistent with the main findings.

By addressing the heterogeneity of labor market responses and emphasizing the role of market size, this paper advances the understanding of how minimum wage policies affect employment and establishments. Our findings contribute to the growing body of evidence

suggesting that under monopsonistic conditions, minimum wage increases can lead to firm expansion and employment growth in larger, more competitive labor markets. These insights have important implications for policymakers, as they highlight the need to consider local labor market structure when designing wage policies.

2 A Theoretical Model of Heterogeneous Labor Market Responses

In this section, we develop a model that formalizes how heterogeneous labor market conditions, specifically differences in labor supply elasticity, can lead to divergent employment outcomes following a minimum wage increase. Our framework is based on standard monopsony models (Boal and Ransom, 1997; Manning, 2003) while incorporating the size of the local labor market as a determinant of the wage setting environment.

2.1 Basic Monopsonistic Labor Market

Consider a representative fast-food firm operating in a county where the local labor market is characterized by an upward-sloping labor supply curve. Denote the wage rate by w and the level of employment by L. The labor supply function is given by

$$L = S(w), \tag{1}$$

with the property that $S'(w) > 0$, reflecting the foundational idea that a higher wage induces a greater quantity of labor supplied. Inverting this relation yields the inverse supply function, $w = S^{-1}(L)$.

Because the firm faces an upward-sloping labor supply curve, hiring an additional worker requires raising the wage for all workers. Consequently, the total labor cost is

$$TC = w(L) \cdot L, \tag{2}$$

and, following Boal and Ransom (1997), the marginal labor cost (MLC) is

$$MLC = \frac{dTC}{dL} = w(L) + L \cdot \frac{dw}{dL}. \tag{3}$$

Under monopsonistic conditions, since $\frac{dw}{dL} > 0$, it follows that $MLC > w(L)$.

2.2 Role of Labor Supply Elasticity and Market Size

Define the local labor supply elasticity as

$$\epsilon = \frac{d \log L}{d \log w} = \frac{w}{L} \cdot \frac{dL}{dw}. \tag{4}$$

In larger counties, where worker mobility and alternative employment opportunities are greater, the labor supply is typically more elastic (i.e., the supply curve is flatter). In contrast, small counties tend to have a less elastic (steeper) labor supply due to limited labor options.

Rewriting Equation (3) in terms of the elasticity ϵ yields

$$MLC = w \left(1 + \frac{1}{\epsilon}\right). \tag{5}$$

Thus, for a given wage w, a higher elasticity ϵ (as in larger counties) implies that the marginal labor cost is only slightly above the wage, whereas a lower elasticity (as in small counties) results in a higher markup of the marginal cost over the wage.

2.3 Implications of a Binding Minimum Wage

Now consider the imposition of a binding minimum wage $w_{\min}$ such that $w \geq w_{\min}$. In a labor market with a relatively elastic supply, as is typical in larger and medium counties, the term $\frac{1}{\epsilon}$ in Equation (5) is small. This implies that the firm's effective cost of hiring additional labor is close to $w_{\min}$, facilitating a more robust increase in employment. In contrast, in small counties with a relatively inelastic labor supply (lower ϵ), the term $\frac{1}{\epsilon}$ is larger, meaning that even with a binding minimum wage, the marginal labor cost remains substantially above the

wage floor. As a result, the employment response in such markets is muted or mixed.

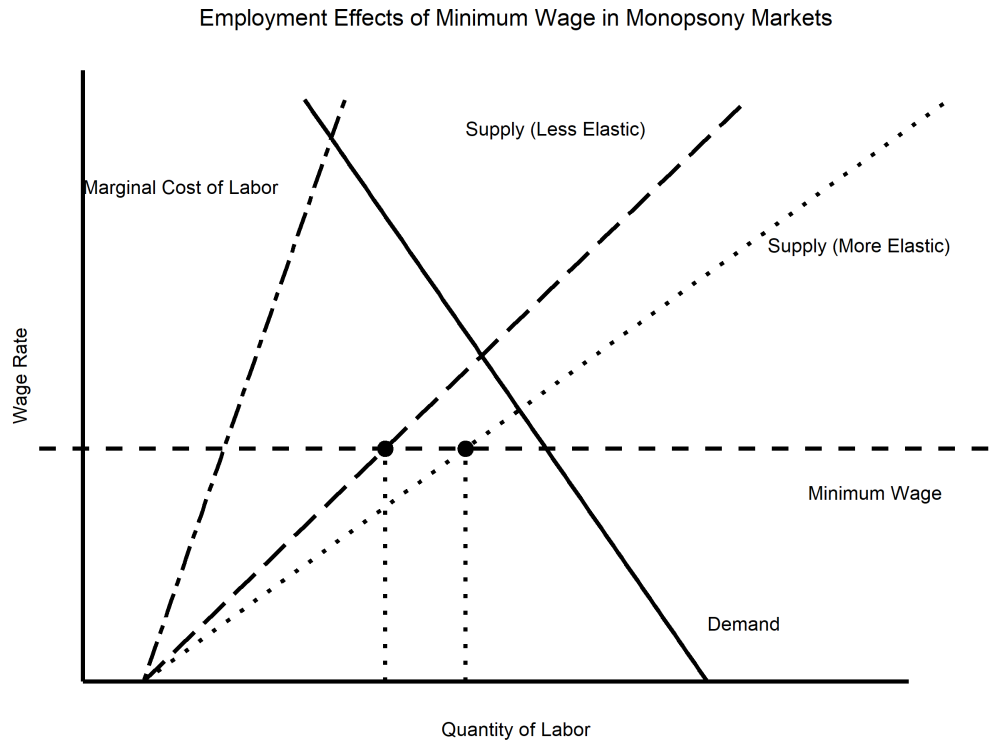


Figure 1: Monopsony and Minimum Wage: Employment Effects Under Different Labor Supply Elasticities

Figure 1 illustrates the effects of a given binding minimum wage in a monopsonistic labor market where the elasticity of the labor supply varies between local labor markets. The steeper labor supply curve, labeled *Supply (Less Elastic)*, represents small counties, where firms have greater wage-setting power due to a more inelastic labor supply. In contrast, the flatter labor supply curve, labeled *Supply (More Elastic)*, corresponds to large and medium counties, where workers are more sensitive to wage changes.

As shown in the figure, when a given binding minimum wage is introduced below the market equilibrium wage, the resulting increase in labor quantity is more pronounced in more elastic labor markets than in less elastic ones. In larger counties, where labor supply is more responsive to wage increases, firms can expand employment significantly when constrained to pay the mandated wage. Conversely, in small counties with a more inelastic labor supply, the increase in labor quantity is relatively muted, as workers are less responsive to wage

changes and firms face higher marginal labor costs. This dynamic helps explain the observed variation in employment responses across different county sizes, reinforcing the role of labor supply elasticity in determining the effects of minimum wage policies.

2.4 Fast Food in Urban Areas and Agglomeration Economy

The assumption that larger counties exhibit a more elastic labor supply is based in several key factors. First, urban areas have a higher population density and a more abundant labor force, which naturally increases the mobility of workers and the availability of jobs. The greater diversity of industries and employment opportunities in large counties allows workers to transition between jobs more easily in response to wage differences, making labor supply more responsive to wage changes.

Second, agglomeration economies further enhance labor market fluidity in urban areas. As firms cluster together, they benefit from shared infrastructure, knowledge spillovers, and a larger pool of potential employees. These dynamics increase labor market efficiency by reducing search costs for both workers and employers, facilitating faster job matching, and improving wage competition. Furthermore, the density of firms fosters greater competition for labor, limiting the wage-setting power of any single employer and increasing overall labor supply elasticity.

In addition, the concentration of fast food establishments in larger cities is influenced by institutional quality and governance. Robles (2015) finds that fast food multinationals tend to expand in cities with stronger governance structures and well-developed supply chains. These factors contribute to a more stable business environment, attract more investment, and reinforce the adaptability of the labor market. In addition, higher levels of urban GDP are associated with greater labor demand and greater flexibility in employment.

Lastly, urban labor markets tend to exhibit higher geographic and occupational mobility. Larger metropolitan areas offer better public transportation networks, shorter job search durations, and access to a wider range of skill enhancement opportunities. As a result, workers in urban environments are more likely to switch sectors or relocate within the city in response to wage fluctuations. Consequently, when a minimum wage increase is introduced in these more elastic labor markets, the marginal labor cost rises less dramatically relative

to the additional labor supplied, potentially leading to employment increases rather than reductions.

In summary, the combination of abundant labor, firm agglomeration, institutional factors, and labor mobility contributes to a more elastic labor supply in larger counties. These conditions shape how urban labor markets respond to minimum wage policies, differentiating them from smaller, less dynamic labor markets.

3 Sample and Data

3.1 Data Sampling and Categorization

The logic for reconstructing the dataset involves a systematic process designed to ensure the validity and robustness of the analysis. We use QCEW and LAUS data, focusing on industries with a high concentration of low-wage workers. Initial data cleaning includes removing counties with disclosure issues and excluding nine California counties that implemented local minimum wage policies.[1] After this step, California counties serve as the treated group, while the donor pool consists of all remaining counties from states that did not experience a minimum wage change between 2009 and 2019.[2]

To address the heterogeneity of economic scales, we classify counties into three size categories: Large, Medium, and Small, based on the number of establishments and employment within each industry of interest. This classification accounts for differences in economic activity across industries and ensures that treated and donor counties are comparable in scale. The size classification thresholds vary by industry to reflect their specific economic distributions. Table 6 in the Appendix provides the detailed thresholds used for each industry.

Categorizing counties by size and selecting the corresponding donor counties based on this classification is a critical feature of our methodological framework. By grouping counties of similar economic scales, we improve the validity of counterfactual estimates generated

[1]Counties in California that implemented local minimum wage policies include Sonoma County, Los Angeles County, San Diego County, Marin County, Contra Costa County, Alameda County, San Francisco County, Santa Clara County, and San Mateo County (Wiltshire et al., Forthcoming).

[2]The donor states include Alabama, Georgia, Idaho, Indiana, Iowa, Kansas, Louisiana, Mississippi, New Hampshire, North Carolina, North Dakota, Oklahoma, Pennsylvania, South Carolina, Tennessee, Texas, Utah, and Wisconsin.

by the synthetic control method (SCM). The treated and donor counties within the same size category exhibit similar economic magnitudes, as shown in Figure 2. This similarity ensures that comparisons are made between economically comparable units, enhancing the robustness of the results.

All variables are normalized relative to their 2014 Q2 values, the final pretreatment quarter, following the approach described in Wiltshire et al. (Forthcoming). The value of each variable in 2014 Q2 is set to 100, and subsequent quarters are expressed as percentages relative to this baseline.[3] Normalizing variables facilitates comparisons across counties and industries while preserving the relative scale of economic changes.

By first selecting donor counties based on raw economic magnitudes and then normalizing variables, we balance the need for realistic counterfactuals with the interpretability of trend-based analyses. This two-step process ensures that both the quality of pre-treatment matches and the post-treatment trends are accurately represented.

3.2 Quarterly Census of Employment and Wages (QCEW)

We use the QCEW dataset as the primary data source for this study. It provides quarterly employment and establishment counts for various industries. This paper focuses on sectors with a high concentration of low-wage workers, including NAICS 722513, 722511, and 44-45. The analysis is restricted to the private sector, as most low-wage workers are employed in this segment (UC Berkeley Labor Center, 2024). California counties are designated as treated units, while the donor pool includes states that have not experienced a minimum wage increase since 2009.[4]

Following the approach outlined in Wiltshire et al. (Forthcoming), we construct quarterly employment by averaging monthly employment figures reported in the QCEW. However, as noted in the same paper, "The QCEW therefore over-weights full-time workers and those who worked the entire quarter. These groups are less likely to be minimum wage workers.

[3]For instance, a normalized employment value of 103 in a given quarter indicates that employment is 103% of its 2014 Q2 value.

[4]The donor states include: Alabama, Georgia, Idaho, Indiana, Iowa, Kansas, Louisiana, Mississippi, New Hampshire, North Carolina, North Dakota, Oklahoma, Pennsylvania, South Carolina, Tennessee, Texas, Utah, and Wisconsin.

As a result, the QCEW may under-estimate minimum wage effects on weekly earnings and employment." Although this limitation may attenuate the estimated impacts on part-time and temporary workers, the QCEW's extensive coverage and detailed industry breakdown remain essential for analyzing local labor market dynamics.

3.3 Local Area Unemployment Statistics (LAUS)

We use the LAUS dataset for county-level unemployment rates and total employment statistics, which are included as covariates in the synthetic control method. These variables account for broader labor market conditions that may influence employment and establishment outcomes, ensuring more accurate counterfactual trends and robust estimates of treatment effects.

3.4 LEADSDEPOSIT Restaurant Data

The LEADSDEPOSIT dataset, as originally provided by LeadsDeposit (2021) and subsequently cleaned and analyzed by Liang and Andris (2021), offers detailed information on restaurants in California. In particular, the dataset includes the percentage of chain restaurants (hereafter, chainness) in each county. We use this measure as an indicator of market concentration, which in turn serves as a proxy for monopsonistic power within the restaurant industry. In our analysis, we examine the correlation between the degree of chainness and the estimated treatment effects of minimum wage increases on employment in fast-food establishments (NAICS 722513).

4 Method

4.1 Construction of Donor Pool

The selection of donor pools is a critical step in the synthetic control method, as emphasized by Abadie (2021). Ensuring that donor units closely resemble treated units is essential to minimize bias and enhance the validity of counterfactual analysis. In our study, donor counties are preselected within each size category (Large, Medium, or Small) based on their

similarity to the treated counties. This stratified approach is motivated by the assumption that counties of different sizes exhibit different labor market structures and dynamics. Given this assumption, matching counties within each size category allows for a more precise estimation of the heterogeneous effects of minimum wage policies.

In later robustness checks, we expand our donor pools to include counties beyond the initial size categories to evaluate the sensitivity of our results. For example, Wiltshire et al. (Forthcoming) classified all counties with more than 5,000 employees as "Large" counties and used this aggregated classification to select donor units. Although this approach was appropriate for their analysis, which did not focus on exploring heterogeneous effects, it would not align with our research objectives. For our study, disaggregating counties by size is necessary to investigate how minimum wage policies affect labor markets with varying economic structures. This methodological distinction highlights the importance of stratified donor pool selection in our analysis.

To determine the similarity between treated and potential donor counties, we calculate the Euclidean distance using two key variables: the logarithm of quarterly average employment levels and the logarithm of quarterly establishment counts. The Euclidean distance is computed as follows:

$$distance = \sqrt{(\ln(emplvl_d) - \ln(emplvl_t))^2 + (\ln(estabs_d) - \ln(estabs_t))^2} \tag{6}$$

Here, $emplvl_d$ and $emplvl_t$ denote the average quarterly employment levels of a donor county (d) and a treated unit (t), respectively, while $estabs_d$ and $estabs_t$ represent the quarterly establishment counts of the donor county (d) and the treated unit (t), respectively. The use of logarithmic transformations ensures that proportional differences are captured, allowing more accurate comparisons between donor and treated counties.

For each treated county, we select nine donor counties for the Large size category, six donor counties for the Medium category, and five donor counties for the Small category. This choice ensures two critical conditions. First, it maintains the similarity of the distributions of donor and treated counties within each size category, improving the validity of counterfactual estimates. Second, it avoids overfitting by limiting the donor pool to a computationally

reasonable size, while still preserving the diversity and representativeness of the donor units. As shown in Figure 2, the treated and donor counties demonstrate similar scales and distributions in terms of establishment counts and employment levels. This comparability underscores the robustness of the donor pool selection process. A comprehensive overview of the treated units and their corresponding donor counties for NAICS 722513 is provided in Table 7.

Table 1: Number of Treated and Donor Units by Industry and County Size

Industry (NAICS)	**County Size**	**Treated Units**	**Donor Units**
Limited-Service Restaurants (722513)	Large	7	22
	Medium	7	35
	Small	25	95
Full-Service Restaurants (722511)	Large	5	23
	Medium	10	32
	Small	25	85
Retail Trade (44-45)	Large	7	30
	Medium	9	39
	Small	27	155

Note: The table presents the number of treated counties and their corresponding donor pools by industry and county size. Industries are classified under the North American Industry Classification System (NAICS).

4.2 Synthetic Control Method

We implement the Synthetic Control Method (SCM) following the framework outlined in Abadie et al. (2010), Alberto Abadie (2015), and Abadie (2021), adopting the mathematical formulation of the stacked synthetic control method from Wiltshire et al. (Forthcoming) and Wiltshire (2025). However, our approach fundamentally differs from Wiltshire et al. (Forthcoming) in that we explicitly account for the heterogeneous effects of minimum wage

policies by stratifying counties into three size categories, Large, Medium, and Small, before estimating synthetic controls. In contrast, the main results in Wiltshire et al. (Forthcoming) apply SCM by combining all donor counties, using a sample of 22 treated counties, 16 in California and 6 in New York, all of which have at least 5,000 employees in NAICS 722. Their donor pool consists of 122 counties that meet the same employment threshold in states that did not experience a change in minimum wage since 2009. Although this approach ensures that treated and donor counties operate in sufficiently large labor markets, it does not account for potential structural differences in labor market responses across different county sizes. Our stratified approach, on the contrary, explicitly categorizes counties into Large, Medium, and Small groups before estimating synthetic controls. This refinement ensures that synthetic controls are constructed using donor units that are more comparable in terms of labor market conditions, leading to more precise and meaningful estimates of treatment effects.

The analysis spans from 2009 Q1 (event quarter 1) to 2019 Q4 (event quarter 44). The pre-treatment period is defined as event quarters 1 to 22[5], with event quarters 1 to 16[6] serving as the optimization window. During this period, we determine synthetic control weights by matching the treated unit with a weighted combination of donor units that best resemble it in terms of key predictors of the labor market (Abadie and Gardeazabal, 2003). Event quarters 17 to 22 serve as a validation period, allowing us to assess the robustness of the synthetic control fit. A well-specified SCM should demonstrate a close pre-treatment fit during this interval, even though no predictors are explicitly matched (Wiltshire et al., Forthcoming).

We use the `allsynth` Stata package (Wiltshire, 2025) to perform placebo tests and estimate confidence intervals, following the statistical inference procedures outlined in Wiltshire et al. (Forthcoming). The package provides robust inferential techniques, including RMSPE-ranked p-values and placebo-variance-based 95% confidence intervals. These methods follow the placebo variance estimation algorithm developed in Arkhangelsky et al. (2021), which adjusts for uncertainty in treatment effect estimates by using the distribution of placebo

[5] 2009 Q1 to 2014 Q2

[6] 2009 Q1 to 2012 Q4

gaps between untreated units. More details on the placebo test design and the construction of confidence intervals can be found in Wiltshire et al. (Forthcoming) and Wiltshire (2025). The predictors used in the SCM, summarized in Table 5, vary depending on the estimated outcome variable. For employment, we include quarterly employment, establishment counts, total employment, and unemployment rates. For establishment counts and weekly wages, we adjust the set of predictors accordingly while maintaining the same general approach to selecting relevant labor market characteristics. To ensure comparability between units, all predictors, except unemployment rates, are normalized to their 2014 Q2 levels, a methodological choice aligned with Abadie (2021) and Wiltshire et al. (Forthcoming).

By incorporating a validation period and using `allsynth` for bias-corrected inference, our SCM implementation ensures robust and interpretable treatment effect estimates. Furthermore, by stratifying counties into size categories, we address a key limitation of previous studies, namely the failure to account for structural differences between labor markets of different sizes. This methodological refinement enhances the reliability of our results and provides new insight into the heterogeneous effects of minimum wage policies.

4.3 Difference-in-Differences with multiple time periods

We use the Difference-in-Differences models developed by Callaway and Sant'Anna (2021) as a robustness check for our main method.

5 Results

In this section, we present the findings of our analysis on the impact of minimum wage policies across counties and industries in California. The results are organized into six subsections, focusing on key outcome variables and mechanisms: employment levels, quarterly establishment counts, average weekly wages, monopsony explanation, delayed response from firms, and robustness checks. We hope to highlight the heterogeneous effects of minimum wage policies and provide additional insights into the role of market structure in shaping these outcomes.

5.1 Employment

The impact of minimum wage increases on employment varies considerably across county sizes and industries. In the limited-service restaurant sector (NAICS 722513), employment exhibited steady growth throughout the treatment period (2014 Q3 - 2019 Q4), with the strongest effects observed in large and medium counties. As shown in Figure 3, employment gains were modest after the first increase in minimum wage (2014 Q3-2016 Q3) but became more pronounced after the second increase (2016 Q3–2019 Q4), suggesting a delayed adjustment process. In contrast, small counties experienced weaker or even negative effects, indicating potential constraints in absorbing wage increases.

Figure 5 presents confidence intervals estimated using the Stacked Synthetic Control Method. The results confirm that the employment gains in large and medium counties are statistically significant, while small counties show considerable uncertainty in the estimates. The strengthening of employment effects over time in large and medium counties suggests that firms adapted to higher wage floors, either through productivity adjustments or increased labor demand.

To assess robustness, Figure 4 reports the results of the placebo test comparing the estimated treatment effects with the synthetic control placebo units. The divergence between treated and placebo counties is particularly evident in large and medium counties, strengthening the credibility of the observed employment gains. In contrast, employment trends in small counties remain closer to the placebo estimates, further supporting the interpretation that minimum wage increases had limited or negative employment effects in these areas.

In the full-service restaurant sector (NAICS 722511), the employment effects are smaller but generally positive, particularly in large and small counties. Medium counties show little change over time, suggesting that adjustments in this sector may have been constrained by demand-side factors or firm cost structures.

In the retail trade sector (NAICS 44-45), the effects are mixed. Large and small counties show positive employment effects, while medium counties experience declines, particularly after the second minimum wage increase. These differences highlight the role of industry-specific factors in shaping labor market responses.

Overall, these findings suggest that the effects of minimum wage increases depend on local labor market conditions, with larger counties exhibiting stronger employment gains, while medium and small counties show more varied responses. The results of Limited-Service Restaurants provide strong evidence that increases in minimum wage can lead to gains in employment in competitive labor markets, while heterogeneity in other sectors underscores the importance of firm and industry characteristics in shaping the adjustment process.

Table 2: Average Treatment Effects on Quarterly Employment Level Across Different Periods and County Sizes

NAICS	**County Size**	**2014 Q3-2019 Q4**	**2014 Q3-2016 Q3**	**2016 Q3-2019 Q4**
722513	Overall	3.10	0.90	5.30
	Large	4.13	1.68	6.59
	Medium	2.47	-0.05	5.00
	Small	-0.43	-1.17	0.30
722511	Overall	1.59	1.85	1.32
	Large	1.92	1.83	2.00
	Medium	0.11	1.65	-1.42
	Small	3.09	2.27	3.90
44-45	Overall	1.41	0.81	2.02
	Large	2.27	1.26	3.27
	Medium	-2.03	-1.36	-2.71
	Small	2.80	2.07	3.53

Note: The table presents the average treatment effects estimated using the Stacked Synthetic Control Method. The three periods correspond to: (i) the full treatment period (2014 Q3–2019 Q4), (ii) the period between the first minimum wage increase (2014 Q3) and the second minimum wage increase (2016 Q3), and (iii) the period since the second minimum wage increase (2016 Q3–2019 Q4). Industry codes follow NAICS classifications: 722513 (Limited-Service restaurants), 722511 (Full-Service Restaurants), and 44-45 (Retail Trade).

5.2 Quarterly Establishment Counts

Similarly to employment trends, establishment growth in response to minimum wage increases was most evident in the limited-service restaurant sector (NAICS 722513), particularly after the second minimum wage increase in 2016 Q3. Table 3 shows that the number of fast food establishments increased by 3.39 percentage points over the full treatment period (2014 Q3-2019 Q4), with the most substantial expansion occurring after the second wage hike, when establishments grew by 6.69 percentage points. The effect was particularly strong in large counties, where the increase reached 8.37 percentage points in the post-2016 Q3 period. Figure 6 further illustrates this pattern, showing a widening gap between treated and synthetic control counties over time.

The expansion of fast food establishments following the minimum wage increase aligns with the idea that higher wages make the industry more attractive to workers, enabling firms to recruit more employees and scale operations. As Reich (2024) quoted, fast food firms benefit from having more workers, since "I would rather sell more hamburgers than fewer. If I have more workers I can sell more burgers." This logic suggests that, while the increase in the minimum wage raised labor costs, it also expanded the effective labor supply, facilitating business growth. The ability to hire more workers allowed firms to increase production capacity, reinforcing the positive employment trends observed in Section 4.1.

By contrast, the effects on full-service restaurants (NAICS 722511) and retail trade (NAICS 44-45) were either negative or negligible. The number of full-service restaurants decreased by 2.08 percentage points over the full period, with the largest reductions occurring in large counties (-3.26 percentage points). Retail trade establishments also declined overall, particularly in medium counties, which saw a drop of 2.52 percentage points. Figures 10 and 12 further illustrate these patterns, showing little to no establishment growth in these sectors.

Differences between industries reflect variations in how firms adjust to higher labor costs. Fast food restaurants can expand more easily than full-service restaurants and retail businesses due to lower overhead costs, simpler operations, and a more flexible workforce. Full-service restaurants face higher fixed costs, greater reliance on skilled labor, and more rigid

service requirements, making expansion more challenging in response to rising wages. Similarly, retail businesses may struggle to adapt due to competitive pressures from e-commerce and broader structural changes in the sector.

A key distinction between the fast food industry (NAICS 722513) and other sectors is its reliance on extensive geographic coverage. Unlike full-service restaurants and retail stores, which often prioritize fewer but higher-revenue establishments, fast food chains require a high density of locations to maximize customer volume. Due to this business model, fast food companies already have a lot of experience in opening new locations and can do so efficiently at a relatively low cost. The minimum wage increase may have provided an additional incentive for expansion, as higher wages improved labor availability, allowing firms to scale operations more effectively. This factor likely contributed to the strong establishment growth observed in fast food following the second minimum wage increase, whereas full-service restaurants and the retail trade did not exhibit similar patterns. While this explanation remains theoretical, it presents a plausible mechanism for why minimum wage increases might facilitate expansion in fast food but not in other sectors.

Overall, these results suggest that the minimum wage increase acted as a catalyst for fast food expansion, particularly after 2016 Q3, when employment effects also strengthened. This indicates that higher wages did not constrain growth in the industry but instead supported expansion by making labor markets more favorable to hiring. In contrast, industries with higher entry costs and operational constraints, such as full-service restaurants and retail trade, did not experience comparable establishment growth.

Table 3: Average Treatment Effects on Establishment Counts Across Different Periods and County Sizes

NAICS	County Size	2014 Q3–2019 Q4	2014 Q3–2016 Q3	2016 Q3–2019 Q4
722513	Overall	3.39	0.09	6.69
	Large	4.31	0.25	8.37
	Medium	2.05	0.85	3.25
	Small	1.18	-1.46	3.83
722511	Overall	-2.08	-2.46	-1.71
	Large	-3.26	-3.56	-2.95
	Medium	-1.80	-1.76	-1.84
	Small	1.23	-0.10	2.56
44-45	Overall	-1.06	-1.24	-0.89
	Large	-0.88	-1.17	-0.60
	Medium	-2.52	-2.21	-2.83
	Small	0.39	-0.03	0.81

Note: The table presents the average treatment effects on establishment counts estimated using the Stacked Synthetic Control Method. The three periods correspond to: (i) the full treatment period (2014 Q3–2019 Q4), (ii) the period between the first minimum wage increase (2014 Q3) and the second minimum wage increase (2016 Q3), and (iii) the period since the second minimum wage increase (2016 Q3–2019 Q4). Industry codes follow NAICS classifications: 722513 (Limited-Service Restaurants), 722511 (Full-Service Restaurants), and 44-45 (Retail Trade).

5.3 Average Weekly Wages

The effects of the increase in the minimum wage on average weekly wages confirm that the policy successfully increased wages in the targeted low-wage industries. Table 4 shows that average weekly wages increased in both the limited-service restaurant (NAICS 722513) and

full-service restaurant (NAICS 722511) sectors over the full treatment period (2014 Q3 – 2019 Q4). In the fast food industry, wages rose by 9.33 percentage points overall, with the strongest gains occurring after the second minimum wage increase in 2016 Q3, when wages increased by 12.27 percentage points. Similarly, the full-service restaurant sector experienced a wage increase of 6.15 percentage points over the entire period. These findings confirm that the policy effectively raised wages in both restaurant industries, aligning with its intended goal.

In all three industries, wage increases were consistently the highest in small counties, the lowest in medium counties, and the lowest in large counties. For example, in NAICS 722513, wages in small counties rose by 13.34 percentage points over the full period, compared to 10.29 percentage points in medium counties and 8.09 percentage points in large counties. A similar pattern holds for NAICS 722511, where small counties experienced the largest wage gains (8.60 percentage points), followed by medium counties (8.31 percentage points), and large counties (4.24 percentage points). This trend may reflect differences in initial wage levels across county sizes, as smaller counties may have had lower baseline wages, leading to greater proportional increases following the policy change.

In contrast, the retail trade sector (NAICS 44-45) exhibited a weaker wage response, with an overall decline in large and medium counties and only marginal gains in small counties. Although the reason for this discrepancy remains unclear, it may be influenced by differences in occupational composition and industry cost structures.

Table 4: Average Treatment Effects on Average Weekly Wages Across Different Periods and County Sizes

NAICS	County Size	2014 Q3–2019 Q4	2014 Q3–2016 Q3	2016 Q3–2019 Q4
722513	Overall	9.33	6.40	12.27
	Large	8.09	5.61	10.58
	Medium	10.29	6.93	13.65
	Small	13.34	9.04	17.64
722511	Overall	6.15	3.93	8.37
	Large	4.24	3.13	5.35
	Medium	8.31	4.66	11.96
	Small	8.60	5.27	11.93
44-45	Overall	-1.73	-1.75	-1.71
	Large	-2.48	-2.47	-2.49
	Medium	-1.05	-0.87	-1.23
	Small	0.71	0.25	1.17

Note: The table presents the average treatment effects on weekly wages estimated using the Stacked Synthetic Control Method. The three periods correspond to: (i) the full treatment period (2014 Q3–2019 Q4), (ii) the period between the first minimum wage increase (2014 Q3) and the second minimum wage increase (2016 Q3), and (iii) the period since the second minimum wage increase (2016 Q3–2019 Q4). Industry codes follow NAICS classifications: 722513 (Limited-Service Restaurants), 722511 (Full-Service Restaurants), and 44-45 (Retail Trade).

5.4 Monopsony Explanation

The findings on employment, establishment counts, and wages align closely with the predictions of a monopsonistic labor market. Under monopsony, firms face an upward-sloping labor supply curve, which means that they set wages below the competitive market equilibrium.

A binding minimum wage can correct for this distortion, potentially leading to employment increases rather than the job losses predicted under perfect competition. However, the extent of this adjustment depends on the elasticity of labor supply, which varies across counties of different sizes.

Our results indicate that large and medium counties experienced significant employment and establishment growth in the fast food sector (NAICS 722513) following the minimum wage increase, particularly after the second wage hike in 2016 Q3. This aligns with the theoretical expectation that in more competitive labor markets, where labor supply is more elastic, the increase in the minimum wage reduces monopsony distortions and leads to an expansion in employment. The muted employment response in small counties, despite wage increases, suggests that firms in these areas faced a less elastic labor supply. With a less responsive labor market, wage increases translated more into higher costs rather than increased hiring. The growth trends of the establishment further support this interpretation. In large counties, fast food expansion was particularly strong after Q3 2016, suggesting that the minimum wage not only improved employment conditions, but also encouraged firms to expand. As labor markets became more attractive, firms were able to hire more workers, allowing them to scale operations. This is consistent with the idea that when labor supply is more elastic, firms can expand output without experiencing steep increases in marginal labor costs.

Additional evidence comes from Figure 8, which shows the correlation between the effects of minimum wage on employment and the restaurant chain rate. While the correlation is weak in small and medium counties, it is more pronounced in large counties, suggesting that in more competitive markets, firms with a higher chainness rate were more likely to expand employment in response to the minimum wage. This further supports the view that in larger labor markets, where labor supply is more elastic, the minimum wage increase reduced labor market frictions, allowing firms to increase hiring.

Overall, these findings indicate that the increase in minimum wage acted to reduce monopsony distortions in the labor market, particularly in larger and more competitive counties. However, in less elastic labor markets, the wage increase primarily led to cost adjustments rather than employment growth, consistent with the predictions of a monopsonistic framework.

5.5 Delayed Response from Firms

The observed lag in establishment and employment growth within the Limited-Service Restaurant sector (NAICS 722513) suggests that firms did not immediately react to the initial minimum wage increase in 2014 Q3. Instead, significant expansion commenced following the second wage hike in 2016 Q3. This pattern implies that businesses may have initially perceived the first increase as an isolated event and thus adopted a cautious approach toward expansion. However, the subsequent increase likely signaled a sustained policy trend, prompting firms to adjust their long-term strategies accordingly.

The lack of immediate expansion following the first increase in minimum wage suggests that firms were initially uncertain about whether higher wages would persist. When the second increase occurred, businesses may have reassessed their expectations, recognizing that increasing minimum wages was not a one-time event, but rather part of a broader trend. This shift in perception likely encouraged firms to expand their operations, particularly in the fast food sector, where the ability to scale up depends on a stable and predictable labor market environment.

In summary, the delayed response to fast food establishment growth appears to be driven by firms' expectations about wage policies. While the first increase may have been viewed with caution, the second increase signaled a long-term shift, leading to expansion strategies that aligned with the new wage landscape.

5.6 Robustness Checks

To ensure the reliability of our findings, we conduct two robustness checks.

The first robustness check employs an alternative implementation of the Difference-in-Differences (DiD) method, as proposed by Callaway and Sant'Anna (2021). This approach accounts for multiple time periods and allows for heterogeneity in treatment effects across units and time. The results of this robustness check are presented in Figures 14, 15, and 16, which display the confidence intervals for employment levels, establishment counts, and weekly wages, respectively. Across all three measures, the results from the DiD estimation show no significant differences from the main analysis, reinforcing the consistency of our findings. In particular,

the positive and significant effects observed in large and medium counties remain robust, while the muted employment response in small counties is also replicated. These results confirm that our main estimates are not driven by model-specific assumptions but hold under an alternative empirical approach.

The second robustness check evaluates the sensitivity of our results to variations in the composition of the donor pool. In the main analysis, we match donor counties to treated counties based on county size classifications. In this robustness check, we maintain the classification of treated counties, but adjust the donor pool selection process. Specifically, for large and medium treated counties, we use all large and medium donor counties as the donor pool, rather than selecting donor counties strictly by size. For small treated counties, we randomly select 150 small donor counties from the full set of available small donor counties to serve as the donor pool. This modification ensures that the results are not overly dependent on the specific donor counties used in the main analysis. Figure 17 presents the results of this robustness check, demonstrating that overall employment trends remain largely consistent with the findings of our main results.

6 Conclusion

This study examines the heterogeneous effects of minimum wage increases across county sizes and industries, focusing on the limited-service restaurant (NAICS 722513), full-service restaurant (NAICS 722511), and retail trade (NAICS 44-45) sectors. Using the Stacked Synthetic Control Method (SCM), we find that the minimum wage increase had a positive impact on employment and establishment growth in the fast food sector, particularly in large and medium counties. However, the effects in small counties were muted, with significant wage increases but limited employment expansion, consistent with the predictions of a monopsonistic labor market.

Our findings suggest that labor market size plays a critical role in shaping firm responses to minimum wage policies. Employment responses vary significantly by county size: In large and medium counties, employment in the fast food sector increased following the minimum wage hike, particularly after the second increase in 2016 Q3. In contrast, small counties

did not show meaningful employment gains despite experiencing the largest wage increases. This suggests that in small counties, where labor supply is less elastic and firms exert greater wage-setting power, firms adjusted to the policy by raising wages rather than expanding employment.

A similar pattern emerges in establishment growth. The number of fast food establishments increased substantially in large and medium counties, particularly after 2016 Q3, indicating that firms expanded operations because they perceived the increase in minimum wage as a sustained policy trend. In contrast, small counties saw little establishment growth, suggesting that higher labor costs discouraged new business openings in less competitive labor markets. However, the full-service restaurant and retail trade sectors did not show evidence of expansion, with the number of establishments stagnating or declining, reinforcing the idea that minimum wage effects are industry-specific and depend on business models and cost structures.

The wage responses further support this framework. The fast food sector experienced the largest wage increases among the three industries, although all were characterized by a high concentration of low-wage workers. This suggests that, prior to the policy change, fast food firms exerted significant monopsony power, keeping wages well below competitive equilibrium. The particularly strong wage increases in small counties, where employment did not expand, further support this view: Employers in these areas probably had the most control over wage setting, and the minimum wage increase forced a substantial upward correction without leading to proportional hiring.

These findings provide new evidence in support of monopsony theory by demonstrating that firm responses to minimum wage increases are shaped by local labor market conditions. Under monopsonistic competition, firms that face a more elastic labor supply, such as those in larger counties, respond to wage increases by expanding employment and establishments. In contrast, in less competitive labor markets, where labor supply is inelastic, firms adjust primarily through wage increases rather than hiring. This insight contributes to the broader minimum wage literature by highlighting the importance of heterogeneity in local labor market structures when evaluating policy effects.

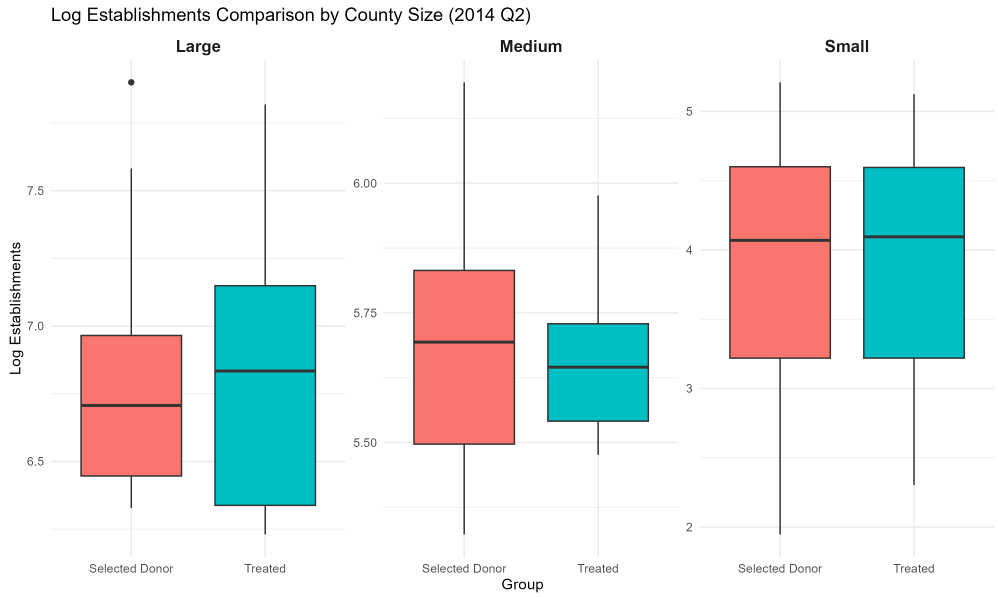
Furthermore, we document a delayed response in employment and establishment growth,

with significant expansion occurring only after the second minimum wage increase in 2016 Q3. This suggests that firms initially viewed the first increase as a one-time event but revised their expectations after the second increase, adjusting their long-term strategies accordingly. Robustness checks using an alternative Difference-in-Differences approach and a modified donor pool selection confirm the stability of our findings. The consistency of results across methodologies reinforces our conclusion that the observed labor market responses reflect structural differences in wage-setting power rather than model-specific assumptions.

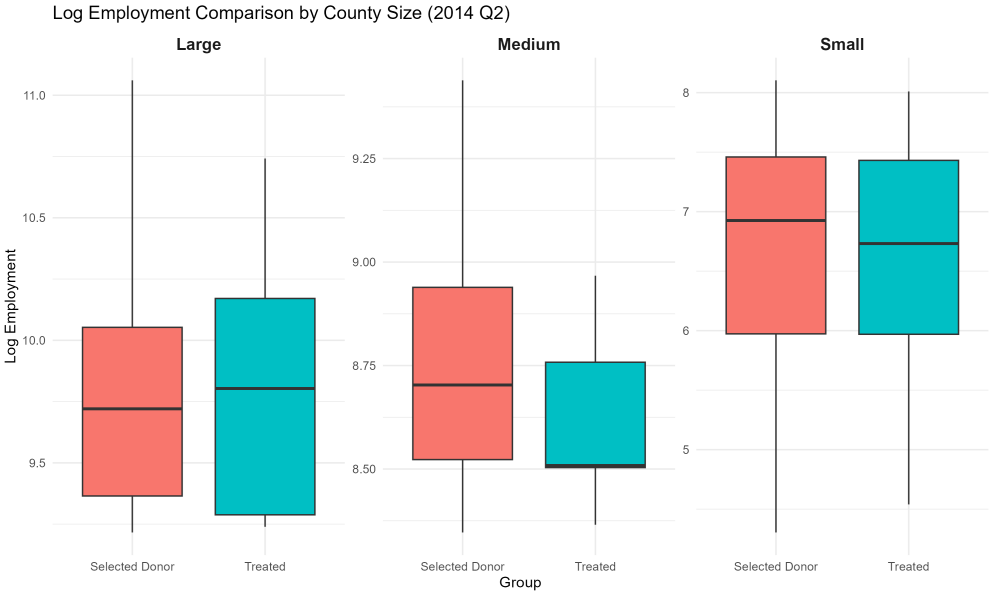
Overall, this study contributes to the minimum wage literature by providing a novel approach that accounts for heterogeneity in local labor market conditions while offering empirical support for monopsony theory. Our results emphasize that minimum wage effects cannot be understood in a one-size-fits-all framework; rather, they depend on the interaction between policy changes and the underlying competitiveness of local labor markets. Future research should continue to explore how firm-level responses to wage policies evolve over time, particularly in the context of changing labor market structures and firm dynamics.

Figure 2: Comparison of Treated and Donor Counties for NAICS 722513

A. Establishment Counts



B. Employment Levels



Note: Panel A shows the distribution of log-transformed establishment counts for donor and treated counties. Panel B presents the distribution of log-transformed employment levels for the same groups.

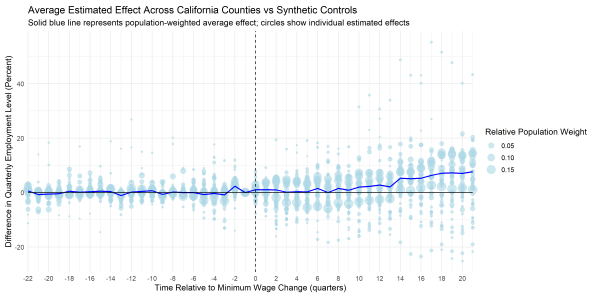
Table 5: Predictors in Synthetic Control Method

Predictor
Normalized Quarterly Employment Level (2009 Q1)
Normalized Quarterly Employment Level (2009 Q2)
Normalized Quarterly Employment Level (2009 Q3)
Normalized Quarterly Employment Level (2009 Q4)
Normalized Quarterly Employment Level (2010 Q1)
Normalized Quarterly Employment Level (2010 Q2)
Normalized Quarterly Employment Level (2010 Q3)
Normalized Quarterly Employment Level (2010 Q4)
Normalized Quarterly Employment Level (2011 Q1)
Normalized Quarterly Employment Level (2011 Q2)
Normalized Quarterly Employment Level (2011 Q3)
Normalized Quarterly Employment Level (2011 Q4)
Normalized Quarterly Employment Level (2012 Q1)
Normalized Quarterly Employment Level (2012 Q2)
Normalized Quarterly Employment Level (2012 Q3)
Normalized Quarterly Employment Level (2012 Q4)
Average Normalized Quarterly Employment Level
Average Normalized Quarterly Establishment Count
Average Total Employment
Average Total Unemployment Rate

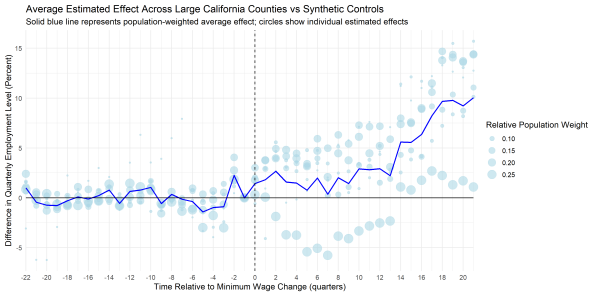
Note: "Average" refers to the mean value of the variable calculated over the optimization window, which spans event quarters 1 to 16 (2009 Q1 to 2012 Q4).

Figure 3: Gap Plots of Treatment Effect on Employment Levels for NAICS 722513

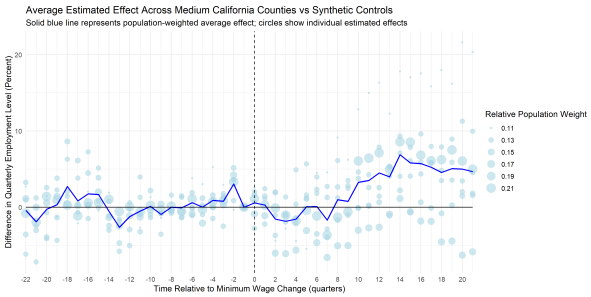
A. All Counties



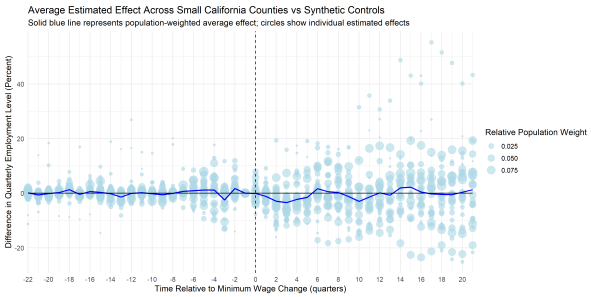
B. Large Counties



C. Medium Counties



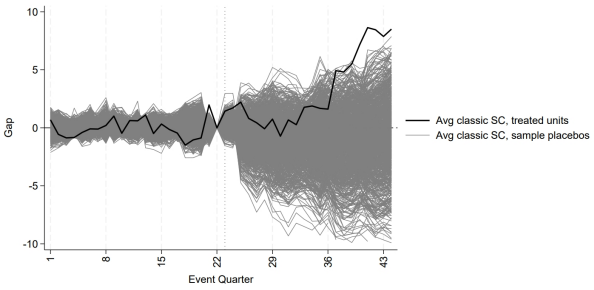
D. Small Counties



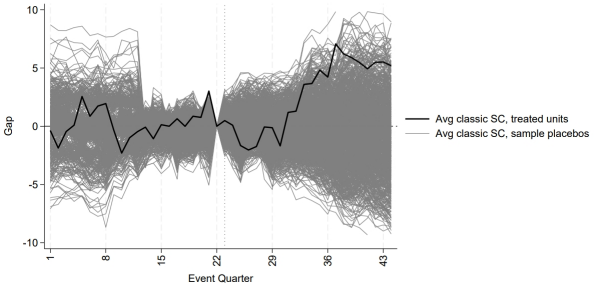
Note: Panels A–D show normalized employment levels for all counties, large counties, medium counties, and small counties, respectively.

Figure 4: Placebo Test: Average Treatment Effects vs. Sampled Placebo Gaps for NAICS 722513

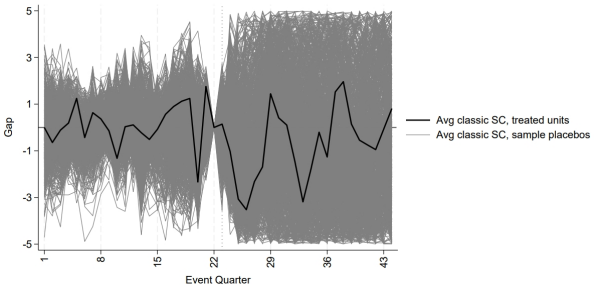
A. Large Counties



B. Medium Counties



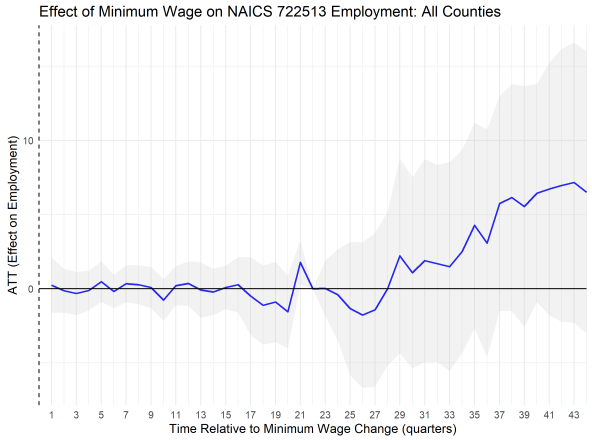
C. Small Counties



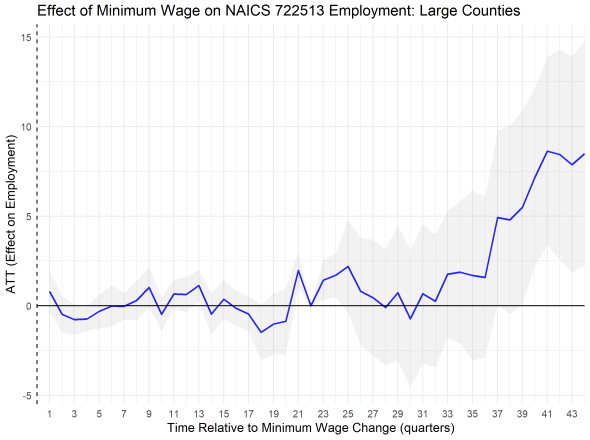
Note: Panels A–C display the placebo test results for large, medium, and small counties, respectively. The dark black line represents the average classic synthetic control result for treated counties, while the grey lines represent the average classic synthetic control results for sampled placebo counties.

Figure 5: Confidence Interval Plots of Treatment Effect on Employment Levels for NAICS 722513 using Stacked Synthetic Control Method

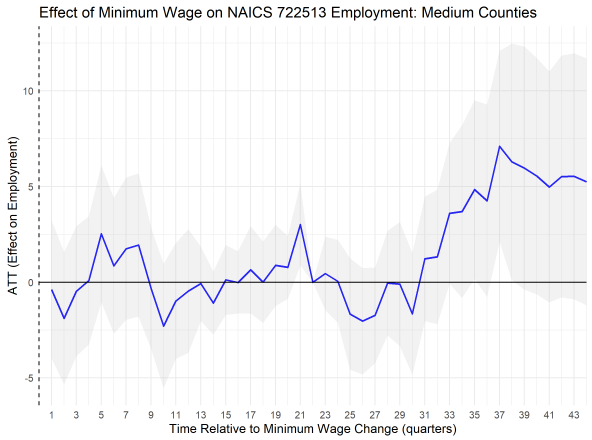
A. All Counties



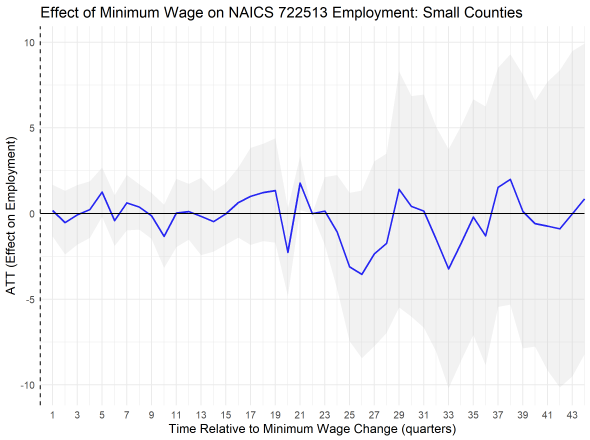
B. Large Counties



C. Medium Counties



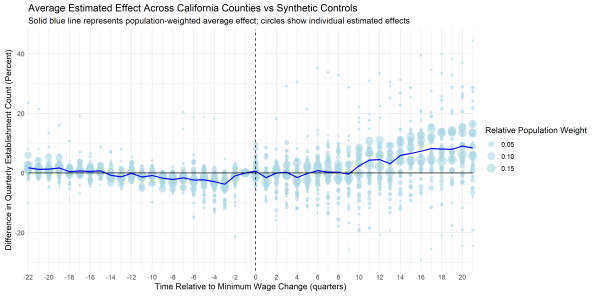
D. Small Counties



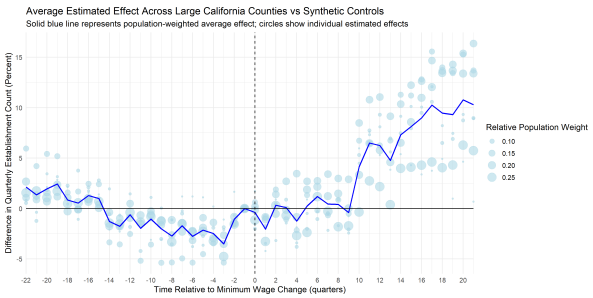
Note: Panels A–D show confidence intervals for all counties, large counties, medium counties, and small counties, respectively. The statistical inference method follows Wiltshire et al. (Forthcoming) and

Figure 6: Gap Plots of Treatment Effect on Quarterly Establishment Counts for NAICS 722513

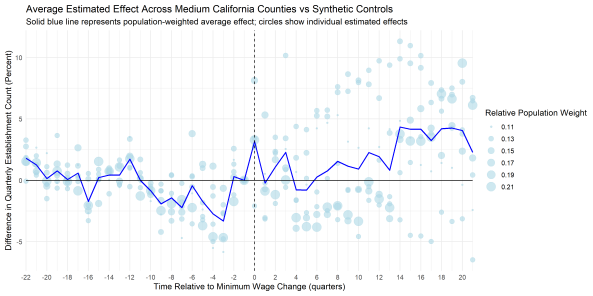
A. All Counties



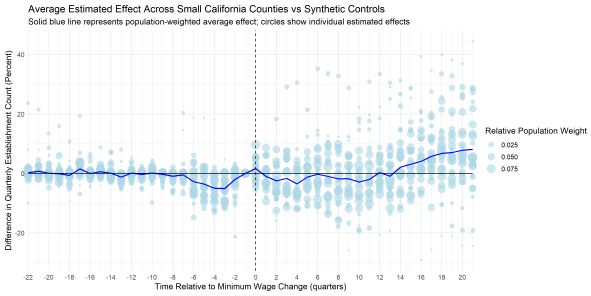
B. Large Counties



C. Medium Counties



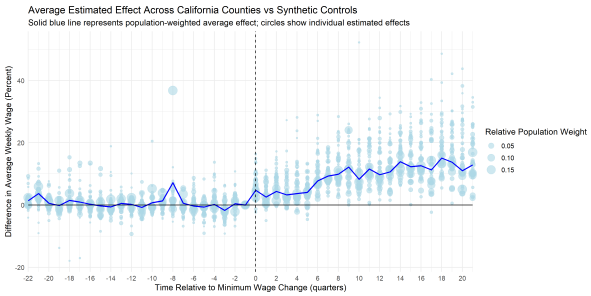
D. Small Counties



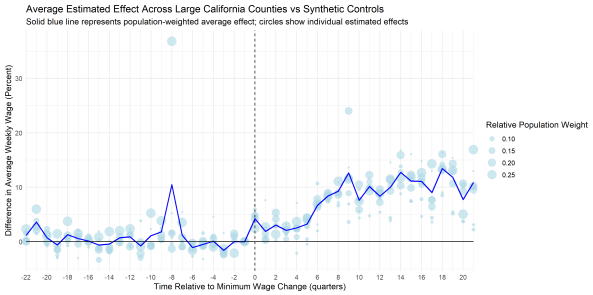
Note: Panels A–D show normalized quarterly establishment counts for all counties, large counties, medium counties, and small counties, respectively.

Figure 7: Gap Plots of Treatment Effect on Average Weekly Wage for NAICS 722513

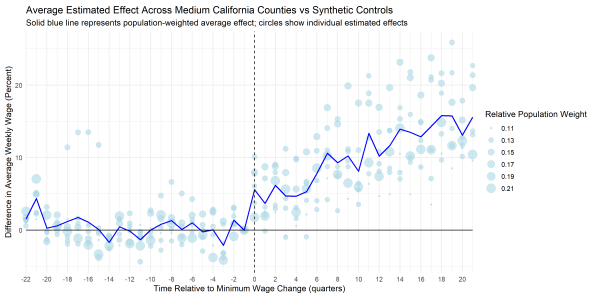
A. All Counties



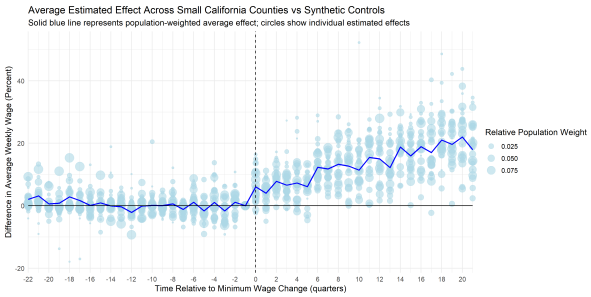
B. Large Counties



C. Medium Counties

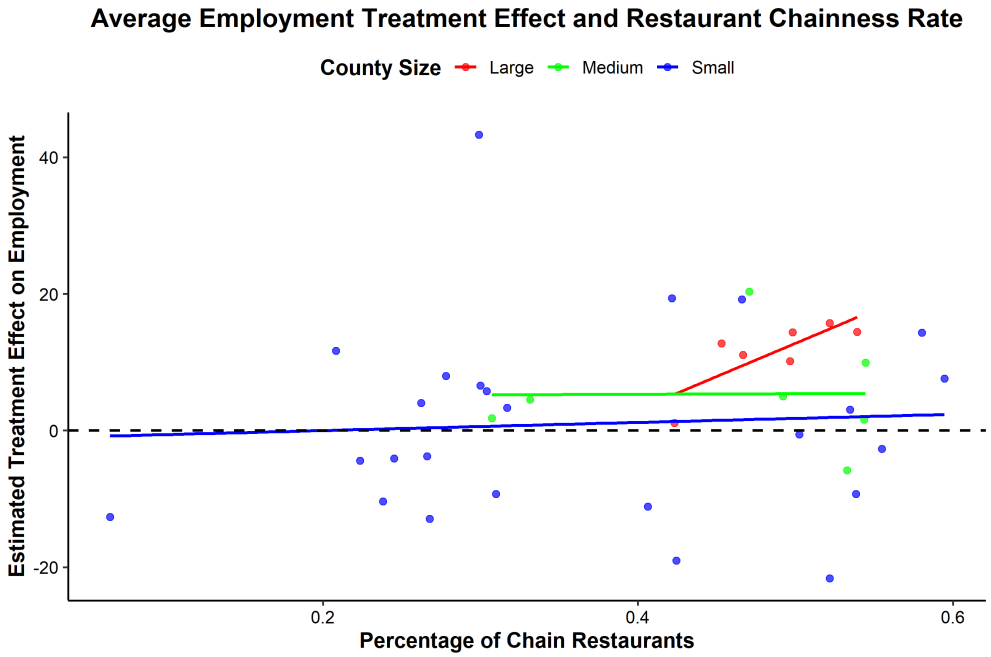


D. Small Counties



Note: Panels A–D show normalized average weekly wage for all counties, large counties, medium counties, and small counties, respectively.

Figure 8: Correlation Between Minimum Wage Effect on Employment and Restaurant Chainness Rate



Note: This figure presents the correlation between the percentage of chain restaurants and the estimated treatment effect of minimum wage increases on employment levels for NAICS 722513. The chainness data is sourced from LeadsDeposit (2021) and Liang and Andris (2021), while the treatment effect is estimated using Stacked Synthetic Control Method at event quarter 44 (2019 Q4). Points are categorized by county size.

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7 Appendix

Table 6: Size Classification Thresholds by Industry (Based on Quarterly Establishment Counts)

Industry	**Large**	**Medium**	**Small**
NAICS 722513	>500	200–500	<200
NAICS 722511	>450	200–450	<200
NAICS 44-45	>1700	800–1700	<800

Table 7: Treated Units and Corresponding Donor Pools by Size Classification for NAICS 722513

Part A: Treated Units

Large
Orange County, CA; San Bernardino County, CA; Riverside County, CA; Sacramento County, CA; Ventura County, CA; Fresno County, CA; Kern County, CA

Medium
San Joaquin County, CA; Stanislaus County, CA; Santa Barbara County, CA; Solano County, CA; Placer County, CA; Tulare County, CA; Monterey County, CA

Small
Santa Cruz County, CA; Butte County, CA; Merced County, CA; Yolo County, CA; Shasta County, CA; Imperial County, CA; Kings County, CA; Napa County, CA; El Dorado County, CA; Humboldt County, CA; Sutter County, CA; Madera County, CA; Mendocino County, CA; Yuba County, CA; Nevada County, CA; San Benito County, CA; Siskiyou County, CA; Lake County, CA; Tuolumne County, CA; Calaveras County, CA; Glenn County, CA; Mono County, CA; Amador County, CA; Del Norte County, CA; Trinity County, CA

Part B: Donor Pools

Large
Harris County, TX; Dallas County, TX; Bexar County, TX; Tarrant County, TX; Travis County, TX; Fulton County, GA; Salt Lake County, UT; Wake County, NC; Marion County, IN; Mecklenburg County, NC; Gwinnett County, GA; Jefferson County, AL; Tulsa County, OK; Milwaukee County, WI; Cobb County, GA; Davidson County, TN

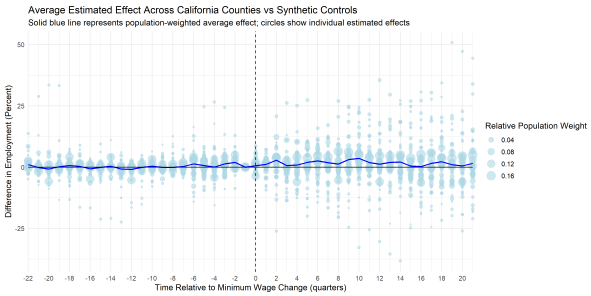
Medium
El Paso County, TX; Guilford County, NC; Johnson County, KS; Greenville County, SC; Fort Bend County, TX; Knox County, TN; East Baton Rouge Parish, LA; Dane County, WI; Jefferson Parish, LA; Nueces County, TX; Madison County, AL; Polk County, IA; Mobile County, AL; Charleston County, SC; Williamson County, TX; Richland County, SC; Hamilton County, TN; Ada County, ID; Forsyth County, NC; Horry County, SC; Waukesha County, WI; Allen County, IN; Chatham County, GA; Durham County, NC; Lexington County, SC; York County, PA; Orleans Parish, LA; Lafayette Parish, LA; Hamilton County, IN; Hinds County, MS; Buncombe County, NC; Montgomery County, AL; Westmoreland County, PA

Small
Clayton County, GA; Shelby County, AL; Rankin County, MS; Linn County, IA; Weber County, UT; Baldwin County, AL; Anderson County, SC; Sullivan County, TN; Gregg County, TX; Cumberland County, PA; Scott County, IA; Alamance County, NC; Hendricks County, IN; Johnson County, IN; Lee County, AL; Hall County, GA; Potter County, TX; Washington County, UT; Johnson County, IA; Union County, NC; Cherokee County, GA; Johnston County, NC; Calhoun County, AL; Houston County, AL; Orange County, NC; Canyon County, ID; Comanche County, OK; Dougherty County, GA; Wilson County, TN; Ascension Parish, LA; Washington County, WI; Wayne County, NC; Forrest County, MS; Beaver County, PA; Eau Claire County, WI; Whitfield County, GA; Marshall County, AL; Etowah County, AL; Rowan County, NC; Nash County, NC; Howard County, IN; Franklin County, PA; Cambria County, PA; Centre County, PA; Blair County, PA; Fayette County, PA; Henderson County, NC; Cullman County, AL; Orange County, TX; Maury County, TN; Lebanon County, PA; Lycoming County, PA; Surry County, NC; Moore County, NC; Carteret County, NC; Walworth County, WI; Rockingham County, NC; Limestone County, AL; Colbert County, AL; Caldwell County, NC; Adams County, PA; Clearfield County, PA; Indiana County, PA; Cheshire County, NH; Columbus County, NC; Crawford County, PA; Pearl River County, MS; Manitowoc County, WI; Beaufort County, NC; Washington County, OK; Dodge County, WI; Thomas County, GA; Somerset County, PA; Carroll County, NH; Covington County, AL; Polk County, GA; Montgomery County, IN; Warrick County, IN; Harvey County, KS; Toombs County, GA; Hancock County, MS; Bedford County, PA; Venango County, PA; Latah County, ID; Putnam County, IN; Armstrong County, PA; Franklin County, NC; Sullivan County, NH; Cherokee County, NC; Huntingdon County, PA; Bonner County, ID; Pierce County, WI; Currituck County, NC; Wayne County, PA; Llano County, TX; Osage County, OK; Polk County, WI; Alexander County, NC; Hoke County, NC; Lincoln County, WI; Tioga County, PA; Cherokee County, AL; Humphreys County, TN; Gillespie County, TX; Shelby County, TX; Boone County, IA; Grimes County, TX; Williamsburg County, SC; Frio County, TX; Richland County, ND; Telfair County, GA; Bamberg County, SC; Bienville Parish, LA; Hale County, AL; Bertie County, NC; Lamb County, TX

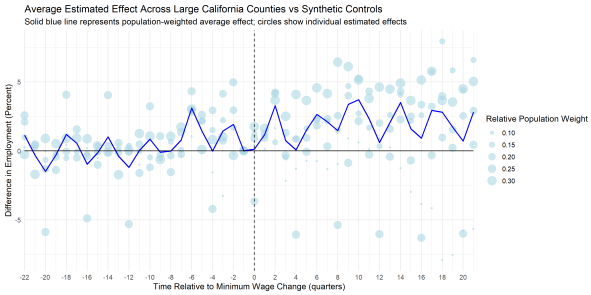
Note: Part A lists the treated units, while Part B lists the corresponding donor pools. Size classifications are determined by the 2014 Quarter 2 average quarterly employment level of NAICS 722513.

Figure 9: Gap Plots of Treatment Effect on Employment Levels for NAICS 722511

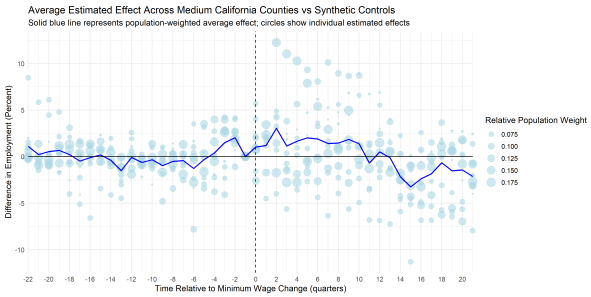
A. All Counties



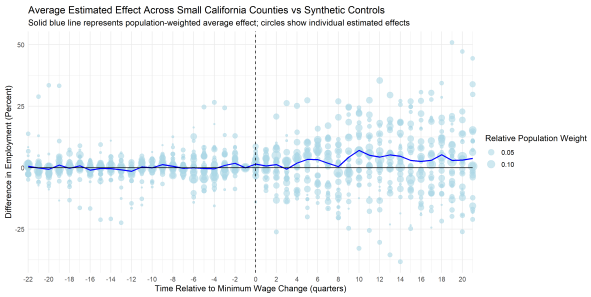
B. Large Counties



C. Medium Counties



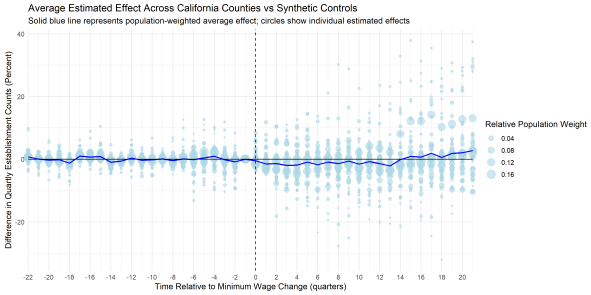
D. Small Counties



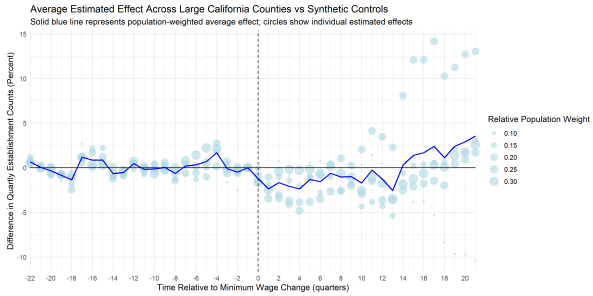
Note: Panels A–D show normalized employment levels for all counties, large counties, medium counties, and small counties, respectively.

Figure 10: Gap Plots of Treatment Effect on Quarterly Establishment Counts for NAICS 722511

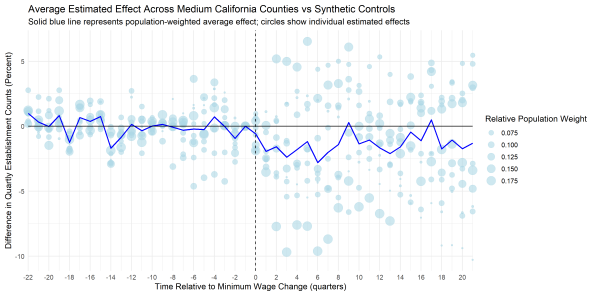
A. All Counties



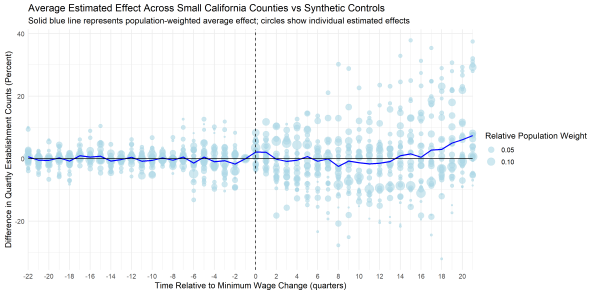
B. Large Counties



C. Medium Counties



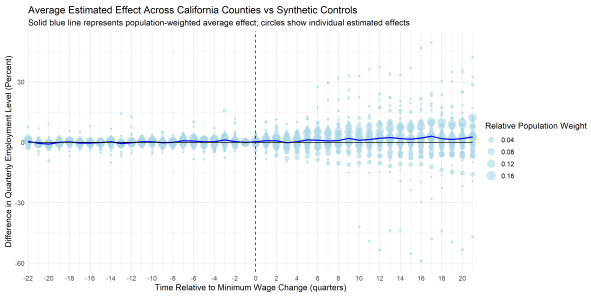
D. Small Counties



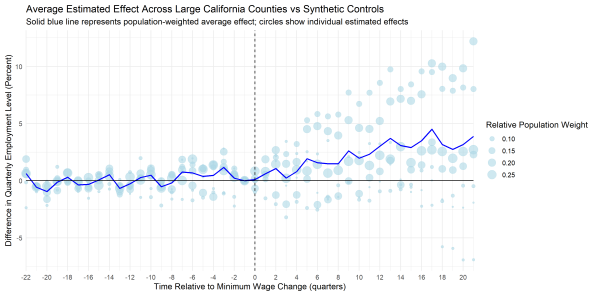
Note: Panels A–D show normalized quarterly establishment counts for all counties, large counties, medium counties, and small counties, respectively.

Figure 11: Gap Plots of Treatment Effect on Employment Levels for NAICS 44-45

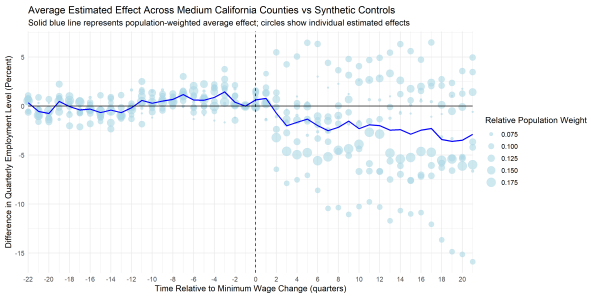
A. All Counties



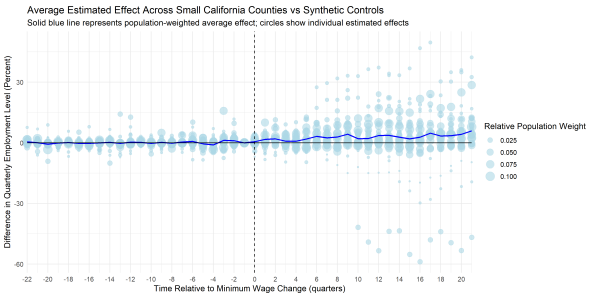
B. Large Counties



C. Medium Counties



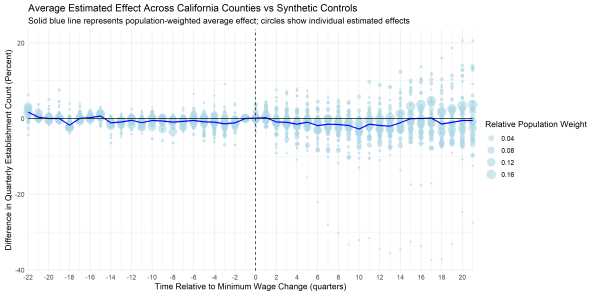
D. Small Counties



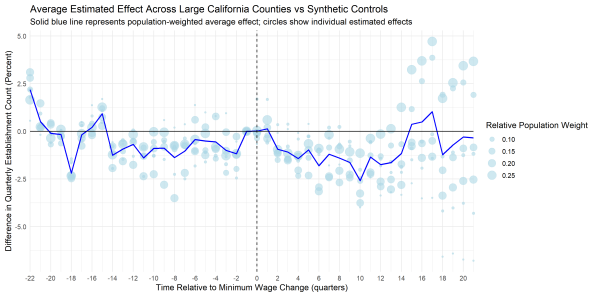
Note: Panels A–D show normalized employment levels for all counties, large counties, medium counties, and small counties, respectively.

Figure 12: Gap Plots of Treatment Effect on Quarterly Establishment Counts for NAICS 44-45

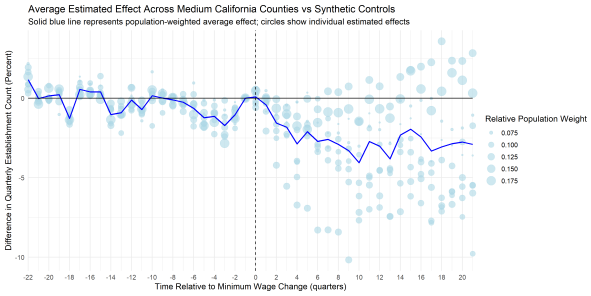
A. All Counties



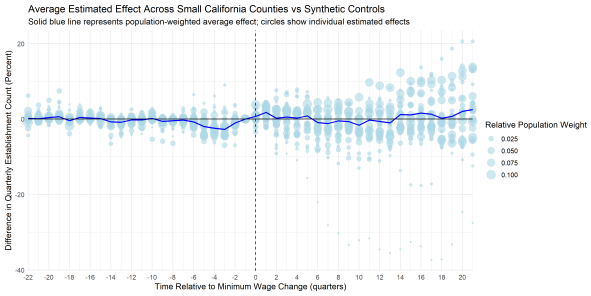
B. Large Counties



C. Medium Counties



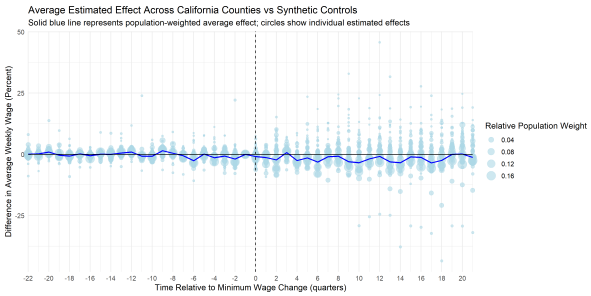
D. Small Counties



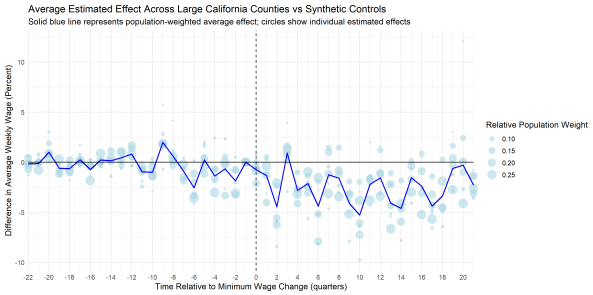
Note: Panels A–D show normalized quarterly establishment counts for all counties, large counties, medium counties, and small counties, respectively.

Figure 13: Gap Plots of Treatment Effect on Average Weekly Wage for NAICS 44-45

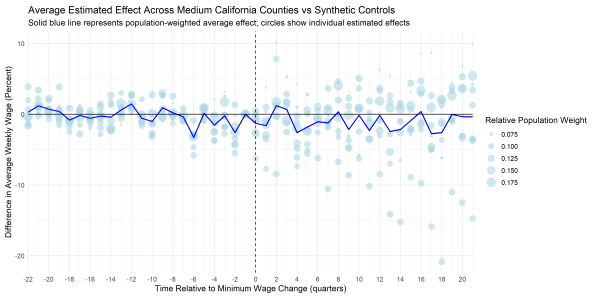
A. All Counties



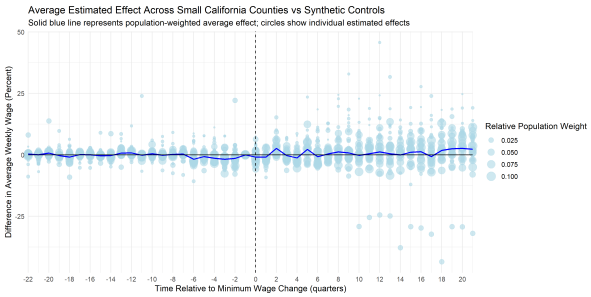
B. Large Counties



C. Medium Counties



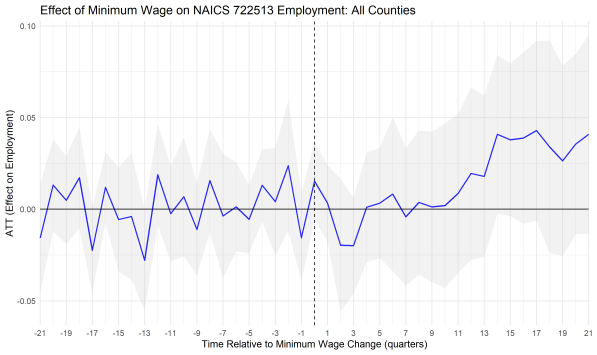
D. Small Counties



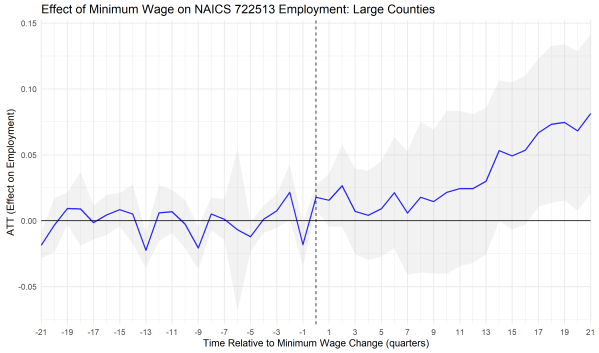
Note: Panels A–D show normalized average weekly wage for all counties, large counties, medium counties, and small counties, respectively.

Figure 14: Confidence Interval Plots for Employment Levels for NAICS 722513 using Difference-in-Differences

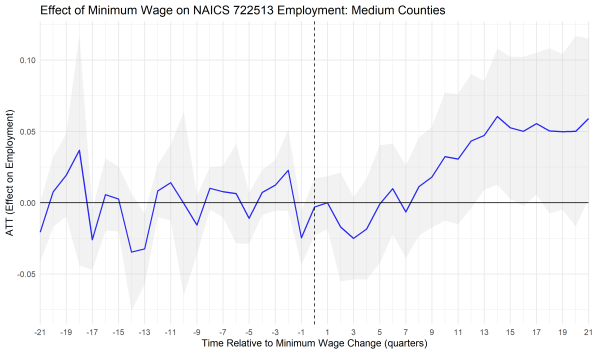
A. All Counties



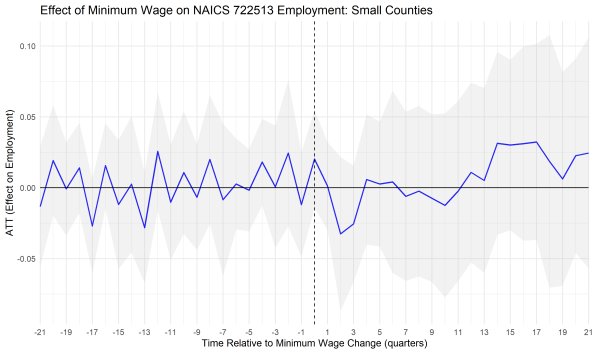
B. Large Counties



C. Medium Counties



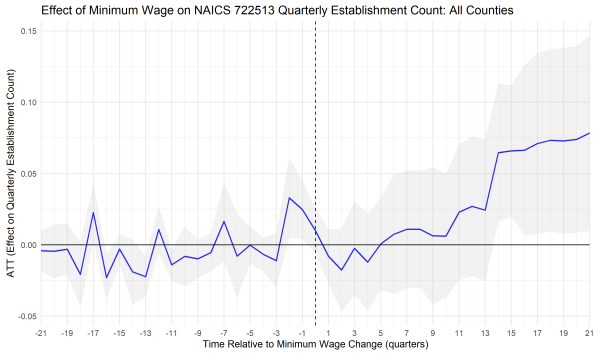
D. Small Counties



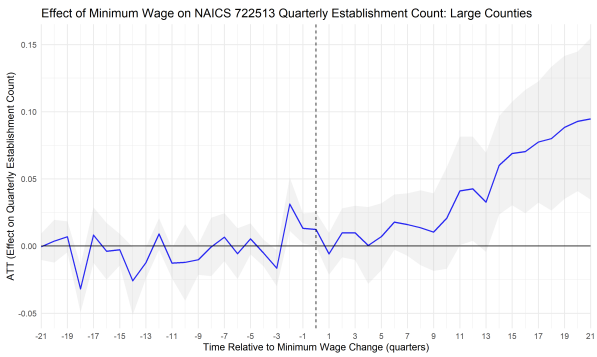
Note: Panels A–D show the confidence intervals for employment levels across the entire period for all counties, large counties, medium counties, and small counties, respectively, using the DID approach developed in Callaway and Sant'Anna (2021).

Figure 15: Confidence Interval Plots for Quarterly Establishment Counts for NAICS 722513 using Difference-in-Differences

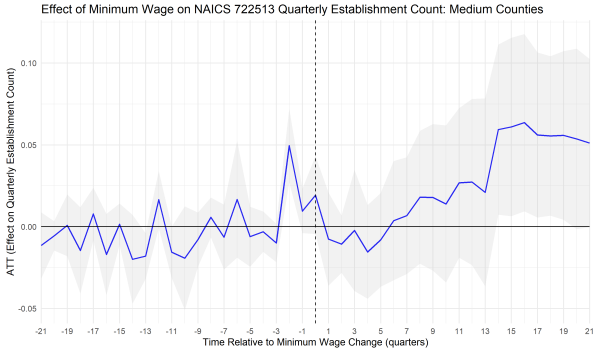
A. All Counties



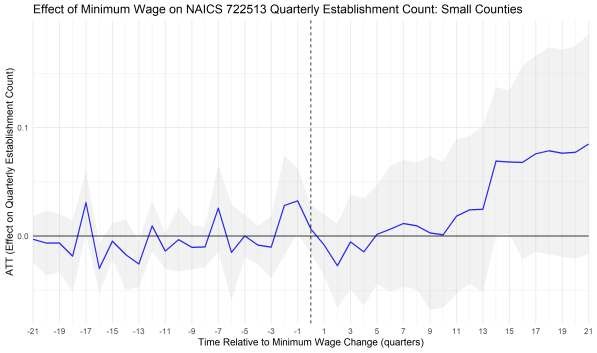
B. Large Counties



C. Medium Counties



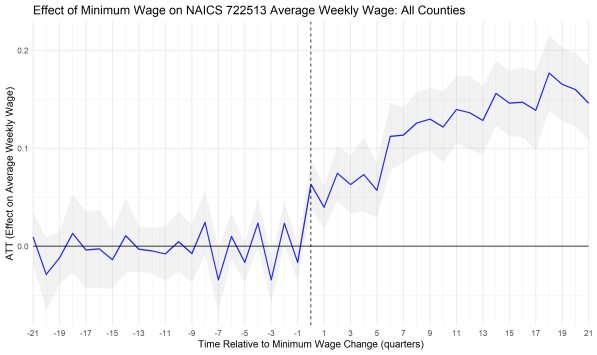
D. Small Counties



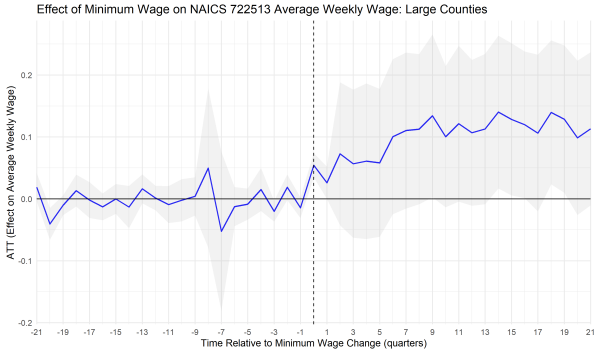
Note: Panels A–D show the confidence intervals for quarterly establishment counts across the entire period for all counties, large counties, medium counties, and small counties, respectively, using the DID approach developed in Callaway and Sant'Anna (2021).

Figure 16: Confidence Interval Plots for Average Weekly Wage for NAICS 722513 using Difference-in-Differences

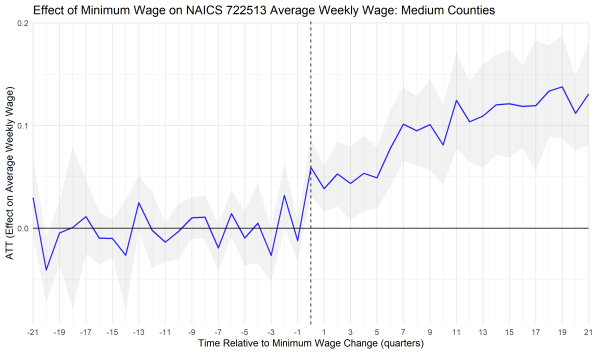
A. All Counties



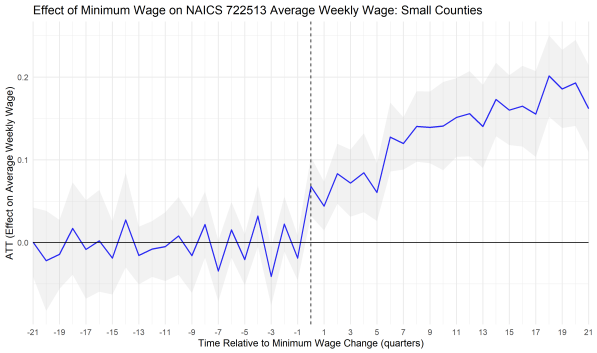
B. Large Counties



C. Medium Counties



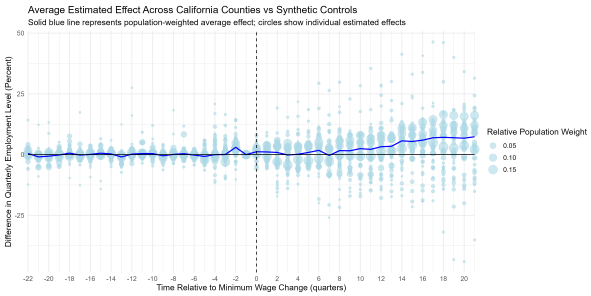
D. Small Counties



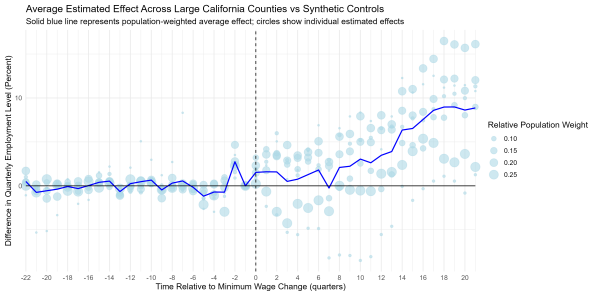
Note: Panels A–D show the confidence intervals for average weekly wage across the entire period for all counties, large counties, medium counties, and small counties, respectively, using the DID approach developed in Callaway and Sant'Anna (2021).

Figure 17: Gap Plots of Treatment Effect on Employment Levels for NAICS 722513

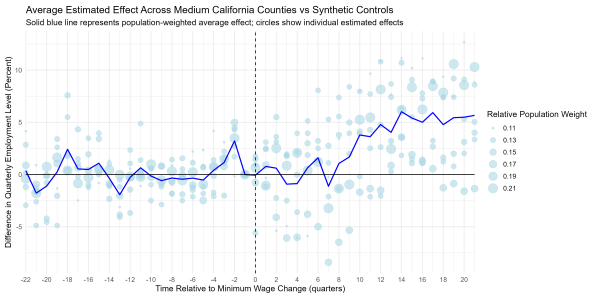
A. All Counties



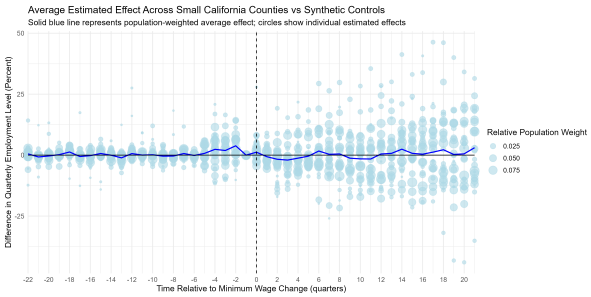
B. Large Counties



C. Medium Counties



D. Small Counties



Note: Panels A–D show normalized employment levels for all counties, large counties, medium counties, and small counties, respectively.